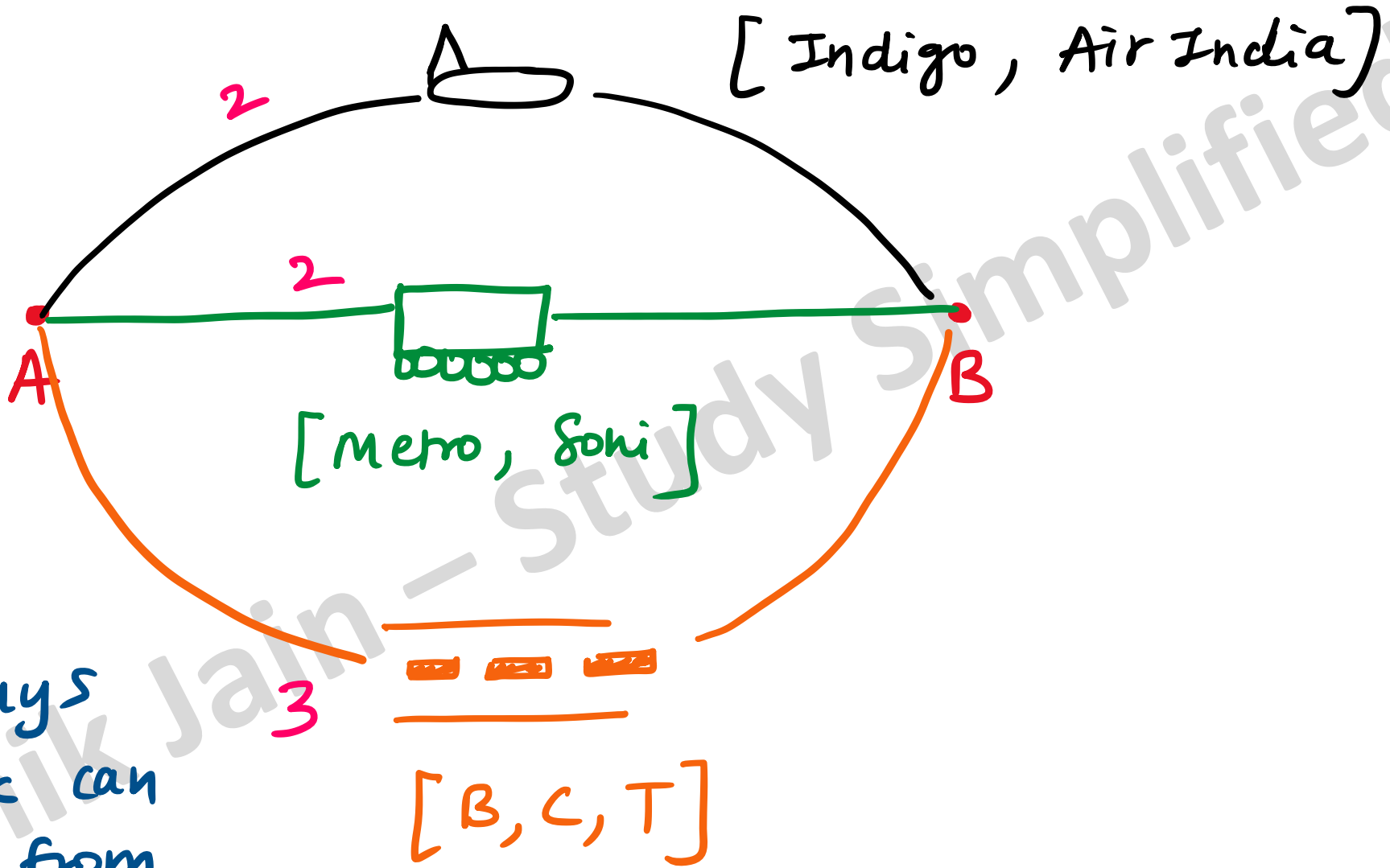


#1



In how
many ways
Vinayak can
travel from
A to B ?

①

$A \rightarrow B$

Indigo (or) A.I (or) M (or) S

(or) B (or) C (or) T

②

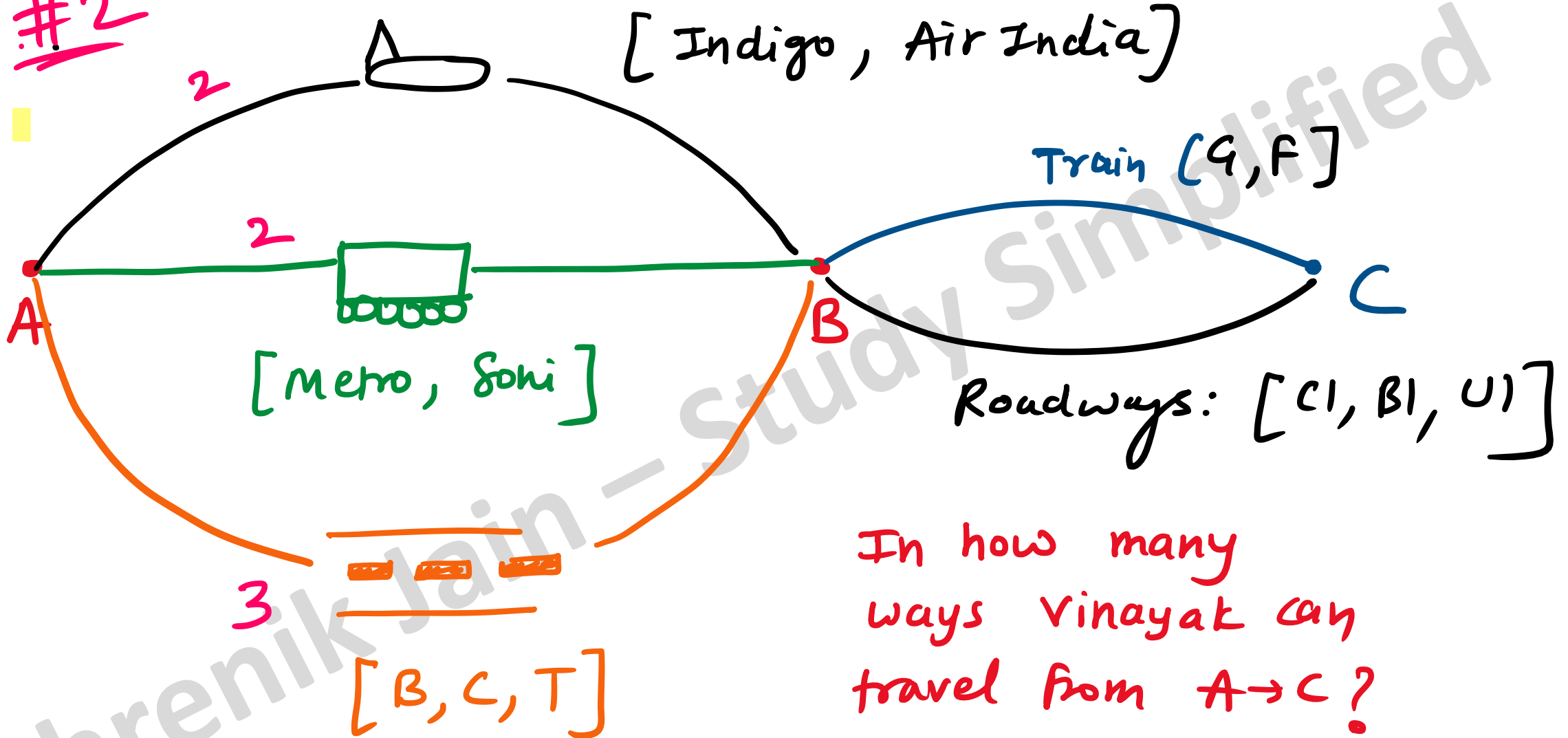
or $\Rightarrow$

(+)

③

$\therefore$ No of ways = 7 //

#2



In how many
ways Vinayak can
travel from A → C?

$A \rightarrow C$

$A \rightarrow B$

(7 ways)

$\nmid$

$B \rightarrow C$

(5 ways)

$$\Rightarrow 7 + 5 = 12^{\times} [wrong]$$

$$\therefore Ans = 7 \times 5 = 35 //$$

or $\Rightarrow$ t
And $\Rightarrow$ x

Shrenik Jain — Study Simplified

. # 3 digit number

 ⁴ ⁴
H.P T.P unit Place

• {1/2/3/4/5}

→ 3 digit no [Repetition]

→ All possible no that can be created is ?

$$\rightarrow \frac{5}{1P} \times \frac{5}{1P} \times \frac{5}{4P} = 125 //$$

$\{1/2/3/4/5\}$

→ 3 digit no [without Repetition]

→ All possible no that can be created is ?

→

$$\frac{5}{HP} \times \frac{4}{TP} \times \frac{3}{4P} = 60 //$$

$\boxed{HP \times TP}$

$\boxed{HP \times 4P}$

$\boxed{TP \times 4P}$

#5★

$$\{0/1/2/3/4/5\}$$

→ 3 digit no

[without repetition]

→ All possible no that can be created is ?

→ $\frac{3}{1P} \times \frac{5}{TP} \times \frac{6}{4P} = 90$ [wrong]

$\boxed{TP \propto 4P \propto 0}$ $\boxed{4P \propto (0-5)}$

NOTE:

① IF 0 comes in picture
then donot go from Right to left
(Directly)

MI

↓
Make cases and then
get answer

↓
fix 0 positions

3 digit no :

$\{0/1/2/3/4/5\}$

case 1
i)

$$\frac{4}{HP} \times \frac{5}{TP}$$

$\boxed{0}$

$$= 20$$

$\boxed{TP \times}$

case 2
ii)

$$\frac{4}{HP}$$

$\boxed{0}$

$$\frac{5}{UP}$$

$$= 20$$

case 3
iii)

$$\frac{3}{HP} \times \frac{4}{TP} \times \frac{5}{UP}$$

$$= 60$$

Ans

$$= 20 + 20 + 60$$

$$= 100 //$$

NOTE :

If 0 comes in picture

(M2)



Just go from left to right

{0/1/2/3/4/5}



Get Ans in
1 step

$$\frac{(5)}{HP} \times \frac{5}{TP} \times \frac{4}{4P} = 100 //$$

(1-5) (HP $\propto$ (0-5)) HP $\propto$ TP $\propto$ (0-5)

3 digit no. {0/1/2/3/4/5}

M3

[Req
= Total
-
unwanted]

$$\begin{array}{c} \underline{6} \\ \text{HP} \\ (0-5) \end{array} \times \begin{array}{c} \underline{5} \\ \text{TP} \\ \text{HPX} \end{array} \times \begin{array}{c} \underline{4} \\ \text{UP} \end{array} = 120 \quad \{2 \text{ 4 3 digit}\}$$

$$\begin{array}{c} \underline{0} \\ \text{TP} \\ (1-5) \end{array} \times \begin{array}{c} \underline{4} \\ \text{UP} \end{array} = 20 \quad \{2 \text{ digit no.}\}$$

$$\therefore \text{Ans} = 120 - 20 = 100 //$$

{0/1/2/3/4/5}

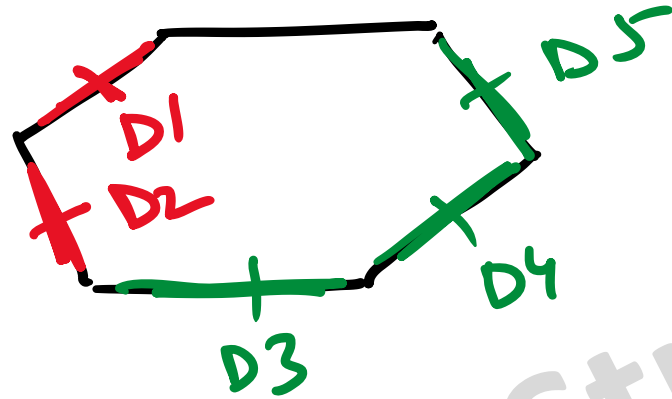
→ 3 digit no [with repetition]

→ All possible no that can be created is ?

$$\rightarrow \frac{5}{\text{HP}} \times \frac{6}{\text{TP}} \times \frac{6}{\text{UP}} = 180 //$$

(0-5) (0-5) (0-5)
02

#

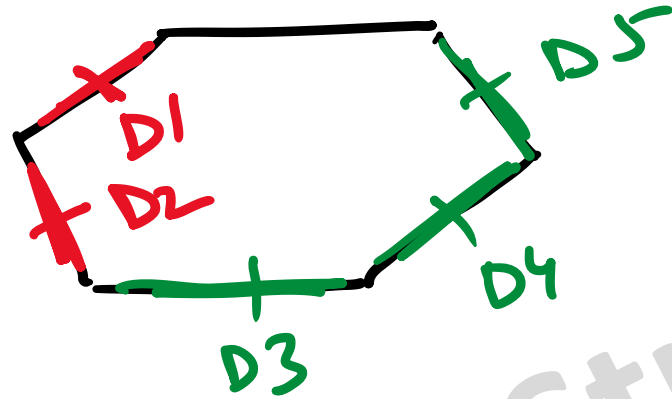


Enter from D_1, D_2
& leaves by D_3, D_4, D_5

m1 :

(D_1, D_3)	(D_2, D_3)
(D_1, D_4)	(D_2, D_4)
(D_1, D_5)	(D_2, D_5)

#



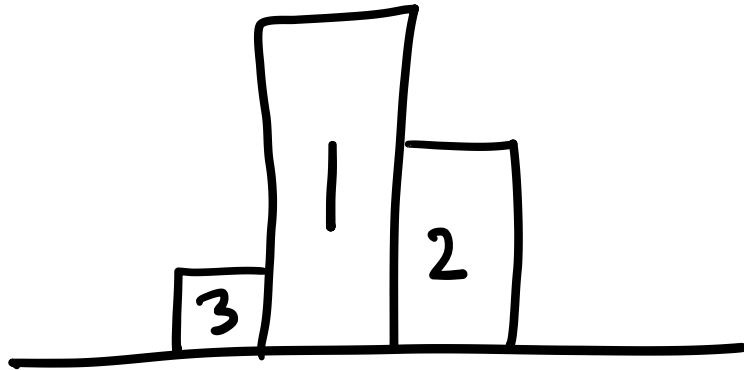
Enter from D₁, D₂
& leaves by D₃, D₄, D₅

M2 :

No of
ways =

$$\begin{array}{c} \text{In} \\ \text{[2]} \end{array} \times \begin{array}{c} \text{out} \\ \text{[3]} \end{array} = 2 \times 3 = 6 //$$

10 students compete in swimming race. In how many ways can they occupy first 3 position.



① without replacement

② $\frac{10}{1^{st}} \times \frac{9}{2^{nd}} \times \frac{8}{3^{rd}} = 720$

How many 4 letter word can be created using
{D, A, U, G, H, T, E, R} if each word include G

(Ans: 840)

① 4 letter word $\rightarrow$ includes G

② G position $\rightarrow$ Not specified

③ Case 1: $\boxed{G} \quad \underline{7} \quad \underline{6} \quad \underline{5} = 210$

Case 2: $\underline{7} \quad \boxed{G} \quad \underline{6} \quad \underline{5} = 210$

Case 3: $\underline{7} \quad \underline{6} \quad \boxed{G} \quad \underline{5} = 210$

Case 4: $\underline{7} \quad \underline{6} \quad \underline{5} \quad \boxed{G} = 210$

Ans
= 840 //

In a test there are **three multiple choice questions** having **four choice each**. Number of sequences in which a student can fail to get all answers correct is

① Total no of sequences
of answers = $\frac{4}{\text{}} \times \frac{4}{\text{}} \times \frac{4}{\text{}}$
= 64

Q1) A **B** C D

Q2) A B **C** D

Q3) **A** B C D

② Only 1 sequence has all correct answer

③ Req = Total - 1
= 64 - 1 = 63 //

∴ # Arrange A, B, C, D in — — — {w/o}

LOGIC 1:

AB	BA	CA	DA
AC	BC	CB	DB
AD	BD	CD	DC

= 12 //

LOGIC 2:

$$\frac{4}{1^{st}} \times \frac{3}{2^{nd}} = 4 \times 3 = 12 //$$

Logic 3:

$$\frac{4 \times 3 \times 2 \times 1}{2 \times 1}$$

$$= \frac{4!}{2!}$$

Arrangement

$${}_n P_r = \frac{n!}{(n-r)!}$$

$$(n \geq r)$$

{Without
Replacement}

• Arrangement of r items out of list of n

$$\rightarrow nP_n = \frac{n!}{0!} = n!$$

$$\rightarrow nP_0 = 1$$

$$\rightarrow nP_1 = n$$

$$\rightarrow nP_{n-1} = n!$$

$$\# n! = n(n-1)!$$

$$\underline{n=1}$$

$$1! = 1(1-1)!$$

$$1! = 0! = 1$$

Arrange A, B, C, D in — — — {with replacement}

LOGIC 1

AA

BA

CA

DA

AB

BB

CB

DB

AC

BC

CC

DC

AD

BD

CD

DD

= 16 //

LOGIC 2

$\frac{4}{1^{\text{st}}}$

$\times \frac{4}{2^{\text{nd}}}$

= 16 //

Logic 3:

n^2

$(n \geq 2)$

[with repetition]

$$A_4 = 4^2 = 16 //$$

$${}_n P_r = \frac{n!}{(n-r)!}$$

→ (w/o)

$$n^r$$

→ (with repetition)

⇒ (Arrangement)

$\therefore \#$ A, B, C, D $\rightsquigarrow$ select 2 letters $\{w/o\}$

LOGIC 1

AB

~~BA~~

~~CA~~

AC

BC

~~CB~~

AD

BD

CD

} 6 ways

LOGIC 2

$${}^4C_2 = \frac{4!}{2!(4-2)!} = \frac{4!}{2!2!} = 6 //$$

selection / Picking

[without Repetition]

→

$${}^n C_r$$

$$= \frac{n!}{(n-r)! r!}$$

$$n \geq r$$

Shrenik Jain — Study Simplified

#

A, B, C, D



select 2 letters

[with repetition]

Logic 1

AA

AB

AC

AD

~~BA~~

BC

BD

BB

CD

CC

DD



10 ways

Logic 2

[with repetition]

$${}^n C_r = \frac{(n+r-1)!}{r! (n-1)!}$$

$(n \geq r)$

$$\frac{(4+2-1)!}{2! (4-1)!}$$

$$= \frac{5!}{2! 3!}$$

$$= \frac{5 \times \cancel{4} \times \overset{2}{\cancel{3}}!}{\cancel{2} \times \cancel{3}!}$$

$$= 10 //$$

[without Repetition] $\rightarrow {}^n C_r = \frac{n!}{(n-r)! r!}$

[with repetition] ${}^n C_r = \frac{(n+r-1)!}{r! (n-1)!}$